Holcombe Department of Electrical and Computer Engineering
Clemson University

Lab Report
Experiment 9 Half Wave Rectifier

ECE 008

ECE-2080

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GTA: Mr. Chang Zou

Instructor: Dr. Bradford White



1) Abstract

In this ninth experiment, the function generator and oscilloscope are used alongside a circuit to create a half wave rectifier. This report will cover the experimental process, and theories to understand and apply half wave rectifiers. This experiment is significant due to the common occurrence of these devices, and how important and irreplaceable they are in the roles they serve.

2) Introduction

The half wave rectifier is a crucial device and commonly found in electrical applications. This experiment will recreate it using a breadboard, signal generator, and view the results from an oscilloscope. The aim of the lab is to create the half wave rectifier but also to continue practicing with the function generator and oscilloscope. Half wave generators can be used to exclusively provide positive voltage and can be found in DC applications where only a small amount of power is necessary such as powering indicator LEDs.

3) Theory and Procedure

This lab only required the usage of two equations for its analysis. These equations both come from oscilloscope readings and are how the data is interpreted. The first equation being RMS, which means root mean square, and is the measurement average peak power.

$$Vrms = \frac{V_m}{2} \quad (1)$$

The V_m represents the max voltage of the waveform. Similarly, the other equation is for peak voltage. This is simply equal to the max voltage of the waveform.

$$Vpeak = V_m$$

These are equivalent due to the half wave rectifier removing the negative voltage making the peak voltage simply equal to the remaining positive peaks. This completed the required equations for the completion of this lab.

Now that the equations have been stated, how the half wave rectifier can be discussed. A half wave rectifier is made up of a few simple components. These being, a diode, and a resistor. The diode serves the purpose of only allowing the positive current through due to its internal electrical barrier that only allows current to flow in one direction. In this experiment, the diode was directionally placed so that only the positive current could flow through. The resistor was added in addition to the diode, to keep voltage and current low enough for the diode to continue operating. This circuit was connected to a function generator which outputs a sinusoidal wave. This wave can be seen displayed on the oscilloscope in figure (1) below.

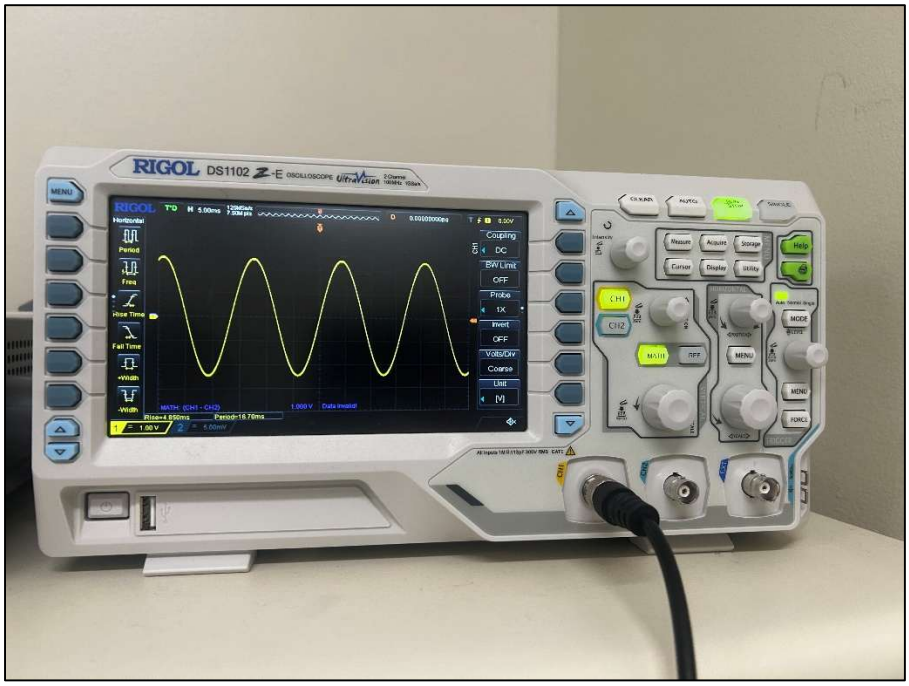


Figure 1: Function Generator Output to Oscilloscope

This sinusoidal wave passes through the circuit, which filters out the negative voltage. The results of this process can be seen below in Figure (2).

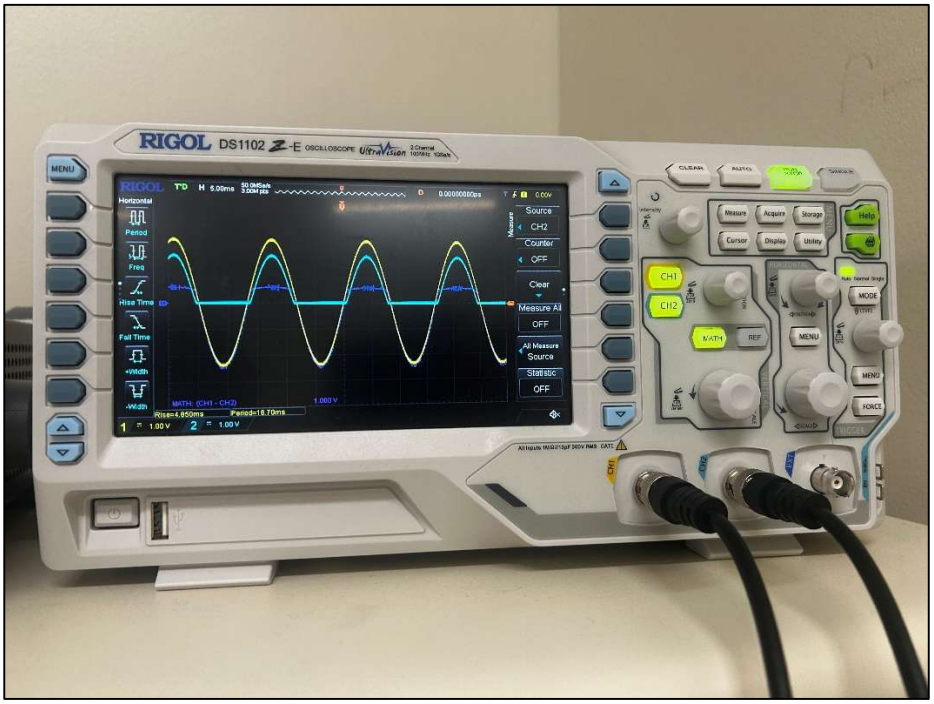


Figure 2: Half Wave Rectifier Displayed to Oscilloscope

I. Lab handout is given.
II. The breadboard is created with the circuit diagram from the handout.
III. The breadboard is connected to the function generator and oscilloscope.
IV. The circuit is double checked, and the table is given power.
V. The function generator is configured for the correct output of 5V at 60Hz
VI. The oscilloscope is turned on and begins collecting data.
VII. Data is recorded onto the lab handout, which can be seen in Figure (3)

4) Discussion of Results

The collection of data was covered in the previous section, and this will cover the results. The results were collected from the oscilloscope, and the ones of note are the Peak Voltage, and the RMS voltage. These are automatically solved by the oscilloscope but could be manually solved using equation (1), and equation (2). These can be seen in Table 1 below. The table is broken down into three sections, channel 1, channel 2, and channel 1 – channel 2. Channel 1 measures the complete sinusoidal wave only being effected by the resistor. Channel 2 is when the measurements are taken across both the diode, as well as the resistor. This is the half wave rectifier. Channel 1 – Channel 2 is when these values are subtracted from each other. These are important to view due to them showing the change in the channels.

Table 1: Experimental Results

CH1 (Voltage at Node A)		CH2 (Voltage at Node B)		CH1-CH2 (Voltage b/w A-B)	
Peak Voltage	RMS Voltage	Peak Voltage	RMS Voltage	Peak Voltage	RMS Voltage
2.44V	1.72V	1.84V	0.863V	0.72V	1.27V

These results are significant because they show that half wave rectifiers work as designed, and that they can be successfully replicated and viewed. Understanding concepts such as these is easier when there is a visual representation. These results are in line with the results from experiment 8, as well as visually correct due to the wave being visibly reformed into a new shape without the negative values.

5) Conclusions

This experiment was a dive into not only half wave rectifiers, but also diodes. Diodes are a building block of complex circuitry, and half wave rectifiers are an excellent usage of them. Although the diodes are important, they were not the focus of this lab. The goal of this experiment was to show the importance of individual components, how they can be used to make circuits, and how the tools in the lab can be used to measure and infer data. This lab is crucial to the skills that students will take away from the course. It provides a summary of the skills learned and is a great conclusion to the lab.

References

[1] A. L. Duke and D. McAuliff, *Electrical Engineering Laboratory I*.

[2] D. K. Freeman *et al.*, "A Sub-millimeter, Inductively Powered Neural Stimulator," *Frontiers in Neuroscience*, vol. 11, Nov. 2017, doi: https://doi.org/10.3389/fnins.2017.00659.

[3] IEEE, "IEEE SA - The IEEE Standards Association - Home," *Ieee.org*, 2019. https://standards.ieee.org/

Appendix A: Technical Standards

[1] IEEE/ANSI 315A-1986, NFPA 70e	(Electrical safety standards)
[2] IEEE 1057-2017	(Testing methods for half wave rectifiers)
[3] IEEE 265-1963	(Measurement and characteristics of diodes)

Appendix B: Documents

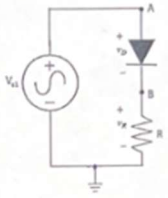
Appdx A
L3 stand IEEE
This Paper
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Abstract
Overview
conclusion
Purpose
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results
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conclusion
Theory
Diode
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Equations
① Vrms
② peak

Laboratory Handout – ECE 2080, Experiment 10

Diode Circuits and Half Wave Rectifier

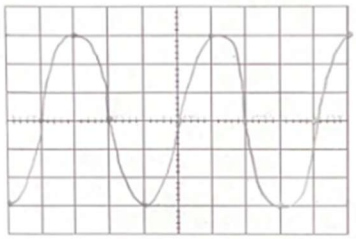
Student Name: Will Meany

1. Connect the following circuit. R = 10 kΩ. Diode is 1N4001 or 1N4004.



2. You will use a function generator to supply V_{s1} with the following,
   a) Select sinusoidal waveform.
   b) Adjust the output voltage to 5 V_{pp} and frequency to 60 Hz. Please note that 60 Hz is the power line frequency in USA.

3. Connect the output of the function generator to the oscilloscope CH1 (Channel 1) at node A by using a BNC cable. Press "Auto" to adjust the display on oscilloscope screen. Draw the waveform in the graph below as it is displayed on the oscilloscope screen and label the waveform for the values on the horizontal and vertical axes. Fill in Table 1.



Voltage axis [V/div]	Time axis [sec/div]	Time period [s]
1 V	5 ms	16.7 ms

Table 1 – Horizontal and Vertical grid for voltage and time axes, respectively.

4. Connect CH2 (Channel 2) of the oscilloscope by using a BNC cable to display the voltage drop across the resistor R_2 at node B.

5. Use the Math function on the oscilloscope to display the voltage drop across the resistor R_1 by selecting A-B, where A=CH1 and B=CH2.

6. Fill-in the following table,

CH1 (voltage at node A)		CH2 (voltage at node B)		CH1-CH2 (voltage b/w A-B)	
Peak Voltage	RMS Voltage	Peak Voltage	RMS Voltage	Peak Voltage	RMS Voltage
2.44 V	1.72 V	1.84 V	0.863 V	0.72 V	1.27 V

Table 2 – Peak voltage and RMS voltage.

TA Signature: _____ Date: 11/20/2025

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History: Prepared by Dr. Hassan Raza on 04/03/2025.

Figure 5: Lab Handout